Compare ⑤ & ⑥

$$\boxed{sY(s) \longrightarrow \frac{2}{T}\,\frac{1-z^{-1}}{1+z^{-1}}\,Y(z)}$$

By bilinear transformation, where T is the sampling time period.

3.b. Design a digital filter by converting the transfer function of analog filter $H(s)$ into digital filter $H(z)$ by using impulse invariant transformation method.

Given $H(s) = \dfrac{s+0.1}{s^2+0.2s+9.01}$

A) Method 1

given $H(s) = \dfrac{s+0.1}{s^2+0.2s+9.01}$

$$= \frac{s+0.1}{s^2+0.2s+0.01+9} = \frac{s+0.1}{(s+0.1)^2+3^2}$$

i.e $H(s) = \dfrac{s+a}{(s+a)^2+b^2}$ where $a=0.1$, $b=3$

using impulse transformation eq.

$$\frac{s+a}{(s+a)^2+b^2} \longrightarrow \frac{1-e^{-aT}(\cos bT)\,z^{-1}}{1-2e^{-aT}(\cos bT)\,z^{-1}+e^{-2aT}z^{-2}}$$

$$\therefore\ H(z) = \frac{1-e^{-0.1T}\cos 3T\,z^{-1}}{1-2e^{-0.1T}\cos 3T\,z^{-1}+e^{-2\times 0.1T}z^{-2}}$$

let $T=1$.

$$H(z) = \frac{1+0.89582z^{-1}}{1+1.7916z^{-1}+0.8187z^{-2}} = \frac{1}{z^{-2}}\left[\frac{1+0.89582z^{-1}}{z^2+1.7916z+0.8187}\right]$$

$$H(z) = \frac{z^2+0.89582z}{z^2+1.7916z+0.8187}$$

Scanned with OKEN Scanner

Scanned with OKEN Scanner

Method 2

given $H(S) = \dfrac{S+0.1}{S+0.2S+9.01}$

$$= \frac{S+0.1}{(S+0.1-3j)(S+0.1+3j)}$$

Partial fractions

$$H(S) = \frac{S+0.1}{(S+0.1-3j)(S+0.1+3j)} = \frac{A}{S+0.1-3j} + \frac{B}{S+0.1+3j}$$

$$S+0.1 = A(S+0.1+3j) + B(S+0.1-3j)$$

let $S = -0.1-3j$

$$\cancel{-0.1} - 3j + \cancel{0.1} = 0 + B(\cancel{-0.1} - 3j + \cancel{0.1} - 3j)$$

$$\cancel{+3j}^{1} = B(\cancel{+6j}^{2})$$

$$B = \frac{1}{2} = 0.5$$

let $S = -0.1+3j$

$$3j = A(6j) + 0$$

$$A = \frac{1}{2} = 0.5$$

$$H(S) = \frac{0.5}{S+0.1-3j} + \frac{0.5}{S+0.1+3j}$$

By impulse invariant transformation, WKT

$$\frac{A_i}{S+P_i} \longrightarrow \frac{A_i}{1-e^{-P_iT}z^{-1}} \quad \& \text{ let } T=1$$

$$H(z) = \frac{0.5}{1-e^{-(0.1-j3)T}z^{-1}} + \frac{0.5}{1-e^{-(0.1+3j)T}z^{-1}}$$

$$= \frac{0.5}{1-e^{-0.1}e^{3j}z^{-1}} + \frac{0.5}{1-e^{-0.1}e^{-j3}z^{-1}}$$

Scanned with OKEN Scanner

Scanned with OKEN Scanner

$$= \frac{0.5\,(1-e^{-0.1}e^{-3j}z^{-1}) + 0.5\,(1-e^{-0.1}e^{3j}z^{-1})}{(1-e^{-0.1}e^{3j}z^{-1})(1-e^{-0.1}e^{-3j}z^{-1})}$$

$$= \frac{0.5 - 0.5\,e^{-0.1}e^{-3j}z^{-1} + 0.5 - 0.5\,e^{-0.1}e^{3j}z^{-1}}{1 - e^{-0.1}e^{3j}z^{-1} - e^{-0.1}e^{3j}z^{-1} + e^{-0.1}e^{3j}\,e^{-0.1}e^{-3j}z^{-2}}$$

$$= \frac{1 - 0.5\,e^{-0.1}z^{-1}(e^{3j} + e^{-3j})}{1 - e^{-0.1}z^{-1}(e^{3j} + e^{-3j}) + e^{-0.2}z^{-2}}$$

$$= \frac{1 - 0.5\,(2\cos 3)\,e^{-0.1}z^{-1}}{1 - e^{-0.1}z^{-1}(2\cos 3) + e^{-0.2}z^{-2}}$$

$$= \frac{1 - \cos 3\; e^{-0.1}z^{-1}}{1 - 2\cos 3\; e^{-0.1}z^{-1} + e^{-0.2}z^{-2}}$$

$$= \frac{1 + 0.8958\,z^{-1}}{1 + 1.7916\,z^{-1} + 0.8187\,z^{-2}}$$

$$\boxed{H(z) = \frac{z^2 + 0.8958\,z}{z^2 + 1.7916\,z + 0.8187}}$$

4.a Compare FIR and IIR filters.

FIR	IIR
→ Designing of FIR filters requires only finite samples of impulse response	→ Designing of IIR filters requires an infinite samples
→ FIR filters are stable	→ IIR filters are not always stable
→ Memory requirement is more	→ Memory requirement is less

Scanned with OKEN Scanner
Scanned with OKEN Scanner

→ Designed for using powerful computational facilities	→ Designed for using hand calculation and tables of analog filter design.
→ poles are fixed at origin	→ poles can be placed anywhere inside the circle
→ Magnitude response shape of FIR filter is controlled by providing greater flexibility	→ Flexibility of IIR filter is less for achieving non standard frequency response
→ A high order transfer function is used to obtain high selectivity	→ A low order transfer function is used to obtain high selectivity.

4.b. Design a digital filter by converting the transfer function of analog filter H(S) into digital filter H(z) by using bilinear transformation method with sampling period T=1 sec Given $H(S) = \frac{2}{s^2+3s+2}$.

Sol:- given $H(S) = \frac{2}{s^2+3s+2}$.

in bilinear transformation to get H(z) $s \rightarrow \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$

$T = 1 sec$

$$s = 2\frac{1-z^{-1}}{1+z^{-1}}$$

$$H(z) = \frac{2}{\left[2\frac{1-z^{-1}}{1+z^{-1}}\right]^2 + 3\left[2\frac{1-z^{-1}}{1+z^{-1}}\right] + 2}$$

$$= \frac{2}{4\frac{(1-z^{-1})^2}{(1+z^{-1})^2} + 6\frac{1-z^{-1}}{(1+z^{-1})} + 2}$$

Scanned with OKEN Scanner

Scanned with OKEN Scanner

$$= \frac{2(1+z^{-1})^2}{4(1-z^{-1})^2 + 6(1-z^{-1})(1+z^{-1}) + 2(1+z^{-1})^2}$$

$$= \frac{2(1+z^{-1})^2}{4(1-z^{-1})^2 + 6(1-z^{-2}) + 2(1+z^{-1})^2}$$

$$= \frac{2(1+z^{-1})^2}{4(1+z^{-2}-2z^{-1}) + 6(1-z^{-2}) + 2(1+z^{-2}+2z^{-1})}$$

$$= \frac{2(1+z^{-1})^2}{4+4z^{-2}-8z^{-1}+6-6z^{-2}+2+2z^{-2}+4z^{-1}}$$

$$= \frac{2(1+z^{-1})^2}{12-4z^{-1}}$$

$$= \frac{\cancel{2}(1+z^{-1})^2}{\underset{2}{\cancel{4}}(3-z^{-1})}$$

$$= \frac{1+z^{-2}+2z^{-1}}{6-2z^{-1}}$$

$$= \frac{1+z^{-2}+2z^{-1}}{z^{-2}[6z^2-2z]}$$

$$= \frac{z^2+1+2z}{6z^2-2z} = \frac{1}{6}\,\frac{z^2+1+2z}{z^2-0.333z}$$

$$= \frac{0.1667z^2+0.333z+0.1667}{z^2-0.333z}$$

Scanned with OKEN Scanner

Scanned with OKEN Scanner

$$\therefore H(z) = \frac{0.1667 z^2 + 0.3332 + 0.1667}{z^2 - 0.3332}.$$

5.a. Write the steps to design digital filter using impulse invariance method.

Steps to design digital filter using impulse invariance method

Step I : For the given specifications find $H_a(s)$, the transfer function of analog filter.

Step II : Select the sampling rate of digital filter, T sec per sample.

Step III : Express the analog filter transfer function as sum of single pole filters

$$H_a(s) = \sum_{k=1}^{N} \frac{C_k}{s - P_k}$$

Step IV : Compute the z-transform of digital filter by using the formula

$$H(z) = \sum_{k=1}^{N} \frac{C_k}{1 - e^{P_k T} z^{-1}}$$

Step V : For high sampling rates use

$$H(z) = \sum_{k=1}^{N} \frac{T C_k}{1 - e^{P_k T} z^{-1}}$$

5.b. Explain frequency transformation in digital domains.

A digital low pass filter can be converted into a digital high pass, band stop, band pass or another digital filter. These transformations are given below.

Scanned with OKEN Scanner

Scanned with OKEN Scanner

Low pass to Low pass:

$$z^{-1} = \frac{z^{-1} - \alpha}{1 - \alpha z^{-1}}$$

where $\alpha = \dfrac{\sin[(\omega_p - \omega'_p)/2]}{\sin[(\omega_p + \omega'_p)/2]} \rightarrow (1)$

ω_p = pass band frequency of low pass filter
ω'_p = pass band frequency of new low pass filter.

Lowpass to highpass

$$z^{-1} = -\left[\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}}\right]$$

where $\alpha = \dfrac{-\cos[(\omega'_p + \omega_p)/2]}{\cos[(\omega'_p - \omega_p)/2]}$

ω_p = passband frequency of lowpass filter
ω'_p = passband frequency of highpass filter.

Lowpass to Bandpass

$$z^{-1} \longrightarrow \frac{-\left(z^{-2} - \frac{2\alpha k}{1+k} z^{-1} + \frac{k-1}{k+1}\right)}{\frac{k-1}{k+1} z^{-2} - \frac{2\alpha k}{k+1} z^{-1} + 1}$$

$$\alpha = \frac{\cos[(\omega_u + \omega_l)/2]}{\cos[(\omega_u - \omega_l)/2]}$$

$$k = \cot\left[\frac{\omega_u - \omega_l}{2}\right] \tan\frac{\omega_p}{2} \rightarrow (3)$$

ω_u = upper cutoff frequency
ω_l = lower cutoff frequency

Scanned with OKEN Scanner
Scanned with OKEN Scanner

Lowpass to Bandstop

$$z^{-1} \longrightarrow \frac{z^{-2} - \frac{2\alpha}{1+k} z^{-1} + \frac{1-k}{1+k}}{\frac{1-k}{1+k} z^{-2} - \frac{2\alpha}{1+k} z^{-1} + 1}$$

$$\alpha = \frac{\cos[(\omega_u + \omega_l)/2]}{\cos[(\omega_u - \omega_l)/2]}$$

$$k = \tan[(\omega_u - \omega_l)/2] \tan \frac{\omega_p}{2}$$

7. Determine $H(z)$ given $H(s) = \frac{1}{s^2+7s+10}$ using
(i) Impulse invariant transformation
(ii) Bilinear transformation method. Assume $T = 0.2$ sec.

A. given $H(s) = \frac{1}{s^2+7s+10}$

(i) Impulse invariant transformation

$$H(s) = \frac{1}{s^2+7s+10} = \frac{1}{s^2+2s+5s+10}$$

$$= \frac{1}{s(s+2)+5(s+2)} = \frac{1}{(s+2)(s+5)}$$

Apply partial fractions

$$\frac{1}{s^2+2s+5s+10} = \frac{A}{s+2} + \frac{B}{s+5}$$

$$1 = A(s+5) + B(s+2)$$

$s = -5 \Rightarrow 1 = 0 - 3B.$

$B = -1/3 = -0.333.$

$s = -2 \Rightarrow 1 = 3A \quad A = 1/3 = 0.333.$

Scanned with OKEN Scanner

Scanned with OKEN Scanner

$$H(s) = \frac{0.333}{s+2} - \frac{0.333}{s+5}$$

By impulse invariant transformation $\omega_k T$

$$\frac{A_i}{s+P_i} \longrightarrow \frac{A_i}{1-e^{-P_i T} z^{-1}}$$

$$H(z) = \frac{0.333}{1-e^{-2T} z^{-1}} - \frac{0.333}{1-e^{-5T} z^{-1}} \qquad T=0.2$$

$$= \frac{0.333}{1-e^{-2(0.2)} z^{-1}} - \frac{0.333}{1-e^{-5(0.2)} z^{-1}}$$

$$= \frac{0.333}{1-0.670 z^{-1}} - \frac{0.333}{1-0.367 z^{-1}}$$

$$= \frac{0.333(1-0.367 z^{-1}) - 0.333(1-0.670 z^{-1})}{(1-0.670 z^{-1})(1-0.367 z^{-1})}$$

$$= \frac{\cancel{0.333} - 0.122 z^{-1} - \cancel{0.333} + 0.223 z^{-1}}{1-0.367 z^{-1} - 0.670 z^{-1} + 0.2458 z^{-2}}$$

$$= \frac{0.101 z^{-1}}{1-1.037 z^{-1} + 0.2458 z^{-2}}$$

$$= \frac{0.101 z^{-1}}{z^{-2}\left[z^{2} - 1.037 z + 0.2458\right]}$$

$$\boxed{H(z) = \frac{0.101 z}{z^{2} - 1.037 z + 0.2458}}$$

Scanned with OKEN Scanner

Scanned with OKEN Scanner

ii, Bilinear transformation

$$H(s) = \frac{1}{s^2 + 7s + 10}$$

$$\left[s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}} \right] \qquad T = 0.2$$

$$H(z) = \frac{1}{\left[\frac{2}{0.2} \frac{1-z^{-1}}{1+z^{-1}}\right]^2 + 7\left[\frac{2}{0.2} \frac{1-z^{-1}}{1+z^{-1}}\right] + 10}$$

$$= \frac{1}{\frac{1}{0.01} \frac{(1-z^{-1})^2}{(1+z^{-1})^2} + \frac{7}{0.1} \frac{1-z^{-1}}{1+z^{-1}} + 10}$$

$$= \frac{1}{\frac{(1-z^{-1})^2 + 7(1-z^{-1})(1+z^{-1})(0.1) + 10(0.01)(1+z^{-1})^2}{0.01(1+z^{-1})^2}}$$

$$= \frac{0.01(1+z^{-1})^2}{(1-z^{-1})^2 + 7(1-z^{-2})(0.1) + 0.1(1+z^{-1})^2}$$

$$= \frac{0.01(1+z^{-1})^2}{1+z^{-2} - 2z^{-1} + 7 - 7z^{-2}(0.1) + 0.1(1+z^{-2}+2z^{-1})}$$

$$= \frac{0.01(1+z^{-1})^2}{1+z^{-2} - 2z^{-1} + 0.7 - 0.7z^{-2} + 0.1 + 0.1z^{-2} + 0.2z^{-1}}$$

$$= \frac{0.01(1+z^{-1})^2}{1.8 + 0.4z^{-2} - 1.8z^{-1}}$$

Scanned with OKEN Scanner

Scanned with OKEN Scanner

$$\boxed{H(z) = \frac{0.01\,(1+z^{-1})^2}{1.8\left[0.22z^{-2}-z^{-1}+1\right]}}$$

8. Determine the cascade form of realization for the system: $y(n)+0.1y(n-1) = 0.2y(n-2)+3x(n) + 3.6x(n-1) + 0.6x(n-2)$

Sol:-

Given $y(n)+0.1y(n-1) = 0.2y(n-2)+3x(n)+3.6x(n-1) + 0.6x(n-2)$

$$y(n)+0.1y(n-1) - 0.2y(n-2) = 3x(n)+3.6x(n-1) + 0.6x(n-2)$$

Cascade form

$$\frac{Y(z)}{X(z)} = ?$$

$$Y(z) + 0.1z^{-1}Y(z) + 0.2z^{-2}Y(z) = 3X(z) + 3.6z^{-1}X(z) + 0.6z^{-2}X(z)$$

$$Y(z)\left[1+0.1z^{-1}+0.2z^{-2}\right] = X(z)\left[3+3.6z^{-1}+0.6z^{-2}\right]$$

$$\frac{Y(z)}{X(z)} = \frac{3+3.6z^{-1}+0.6z^{-2}}{1+0.1z^{-1}+0.2z^{-2}}$$

$$= \frac{3+3z^{-1}+0.6z^{-1}+0.6z^{-2}}{(1+0.5z^{-1})(1-0.4z^{-1})}$$

$$= \frac{3(1+z^{-1})+0.6z^{-1}(1+z^{-1})}{(1+0.5z^{-1})(1-0.4z^{-1})}$$

$$= \frac{(1+z^{-1})(3+0.6z^{-1})}{(1+0.5z^{-1})(1-0.4z^{-1})} = \frac{(3+0.6z^{-1})(1+z^{-1})}{(1+0.5z^{-1})(1-0.4z^{-1})}$$

Scanned with OKEN Scanner

Scanned with OKEN Scanner